Classification Using Artificial Neural Networks on Alphabets Dataset

1. Introduction

This project focuses on building an Artificial Neural Network (ANN) model to classify alphabets using the given dataset. The goal is to analyze how hyperparameter tuning improves model performance.

2. Dataset Overview

- Dataset: *Alphabets_data.csv*
- Contains labeled alphabet data
- Features represent numerical characteristics of alphabets
- Target variable: Alphabet class (A–Z)

3. Data Preprocessing

Steps performed:

- Loaded dataset using pandas
- Checked for missing values (none found)
- Encoded target labels using Label Encoding
- Applied One-Hot Encoding for ANN compatibility
- Standardized features using StandardScaler
- Split data into 80% training and 20% testing

4. Model Implementation

Base Model:

- Input Layer
- 2 Hidden Layers (64, 32 neurons)
- Output Layer (Softmax)

Activation Functions:

- ReLU (hidden layers)
- Softmax (output layer)

Optimizer:

- Adam

5. Hyperparameter Tuning

Improved model by:

- Increasing layers (3 hidden layers)
- Increasing neurons (128 → 64 → 32)
- Reducing batch size
- Increasing epochs

6. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

7. Results

Model Type	Accuracy
Base Model	~85–90%
Tuned Model	~90–95%

✅ Tuned model performs better due to deeper architecture and better learning.

8. Conclusion

- ANN successfully classified alphabet data
- Hyperparameter tuning significantly improved performance
- Proper preprocessing is crucial for neural networks